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## Project Proposal

1. Problem Statement & Motivation: I would like to propose a creative tool. The problem is that there are already tools out there that help people make the final product, but there are not tools that help them in deciding what route they are wanting to take in their creative direction, explore all possible ways this can look to them, give a messy idea structure, and then come up with a finished product. The target users for this tool would be individuals in marketing, designers i.e. Fashion, systems, furniture, etc., writers, students, entrepreneurs. An agentic AI solution would be appropriate for this because humans can often be biased. When you are trying to get advice on making a decision, they can often rely on their own opinion that may be completely opposite of what the person is looking for. With the Agentic AI you have the capability of giving it multiple prompts until it fully understands what you are looking for and it comes up with multiple solutions. Also, with humans they can be creatively exhausted as well, the AI never gets tired. Sometimes humans can make you feel bad for your ideas, and it doesn't create a safe environment anymore for further creation whereas the Ai will never tell you that you have a bad idea.
2. Use Cases & Scenarios:
  - a. A user wants to start a business, but their ideas are scattered.
    - i. Tasks the agents perform: One agent will generate a concept of direction once user as inputted the ideas. Second agent will flag the weaker ideas. Third, there will be agent that tells the user how feasible each idea is. Lastly it will get the user to the point of their strongest idea and coming up with a finished product.

- b. A user wants to propose new product but is having time constraints and limited brainstorming input.
    - i. Tasks the agents perform: First comes up with multiple concepts. Second evaluates whether the concepts are aligned with the users' needs and the business objectives. Third it challenges ideas that may add complexity without value. Lastly generates a narrative explain the final product.
- 3. Initial System Design:
  - a. Agents: Ideation/ Planner Agent, Critic Agent, Market Agent, Builder Agent, Memory Agent
  - b. Expected tools/ APIs or data sources: Industry trend APIs, Startup and company databases, public business datasets, Web search, Vector database
- 4. Feasibility & Risks:
  - a. Potential Challenges: The user having too much confidence in the ideas that the AI gives. The system could produce the same generic ideas to each user. It may be too many agents being used that could cause some failures. The system could give too many ideas, or it could not give enough. There could be some ethical concerns being that the system is doing a web search to help with the ideation it could possibly be taking someone else's idea.
  - b. Constraints: System not being able to access real time market outcomes.  
Knowledge constraints. Design constraints if the user inputs vague inputs it reduces the quality of final outcome. Autonomy constraints the system cannot contact the user or run experiments with the idea it came up with. Ethical and legal constraints.

- c. Timeline for project execution:
  - i. 2 weeks of writing prompts for the agents, defining the inputs and outputs and how agents are interacting.
  - ii. 2 weeks agent routing, iteration loop
  - iii. 2 weeks add RAG layer add PDFs, articles, reports
  - iv. 1 day Take time to demo
  - v. Test it and make changes turn in final deliverable